ZCAS UNIVERSITY

CCS3631 - MOBILE AND WIRELESS FUNDAMENTALS

END OF SEMESTER 1 EXAMINATION

2ND JUNE, 2025

09:00 HRS-12:00HRS

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

(a) (a) Wi-Fi is a wireless networking technology that uses radio waves to provide wireless high-speed Internet and network connections. There have been several Wi-Fi standards released over the years, each with different specifications for data transfer rates, frequency bands, security, and other features. You are working as a network engineer, and you have been tasked to recommend a wireless standard to use in the company's network. To this effect, you are to describe the various characteristics of Wi-Fi standards. Complete the table below to accomplish this task. **[20 Marks]**

WiFi Version	IEEE Standard	Frequency	Speed	Modulation	Security
WiFi 4					
WiFi 5					
WiFi 6					
WiFi 7					

(b) With the aid of a well labeled diagram, describe the Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) protocol in WiFi networks. **[10 Marks]**

(c) With the aid of a well labeled diagram, briefly describe the Request to Send/Clear to Send (RTS/CTS) mechanism and explain how it helps to mitigate the hidden node problem in wireless networks. **[10 Marks]**

(Total: 40 marks)

SECTION B: Attempt any THREE questions in this section.

Question 2

There are different wireless network technologies that integrate with traditional networks. A GSM network is a complex system composed of various subsystems that work together to provide seamless mobile communication services. Consider the GSM subsystems depicted in Figure 1 below.

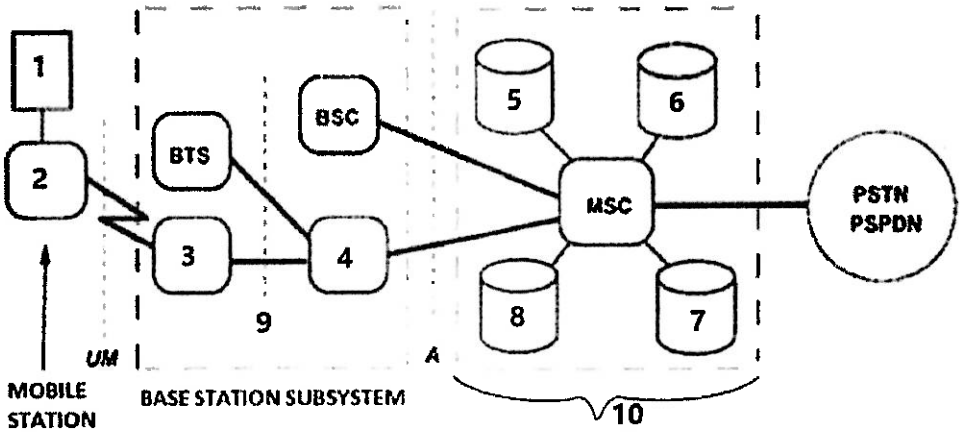


Fig.1. GSM subsystems

i. Label the parts of the GSM subsystem shown 1 to 10. **[5 Marks]**

ii. Explain, with the help of the components in Fig.1, how a SIM is authenticated in the GSM networks to provide network connectivity. **[5 Marks]**

iii. Explain, in tabular form, the differences between 4G and 5G networks. **[10 Marks]**

Question 3

Imagine you are a working for business owner who needs to provide internet access to her employees. She could choose to set up a Wi-Fi network at the offices, or she could use Internet through a cellular network, or a satellite network. As the network engineer, you need to emphasize the importance of understanding the differences between Wi-Fi and cellular networks.

Parameter	**WiFi Network**	**Cellular Network**	**Satellite Network**
Range (Distance)			
Data transfer rates			
Frequency			
Security Protocols			
Example of Standards			

a) Complete the table below to accomplish this task. **[15 Marks]**

b) Explain the major types of frequencies used between UT and Satellite, and between Satellite and Earth station. **[5 Marks]**

Question 4

In cellar networking, it's essential to understand the fundamental concepts of connection transitions. In this context, two primary handover mechanisms exist: Break Before Make (BBM) and Make Before Break (MBB). These mechanisms dictate how a mobile device transitions its connection from one base station to another, impacting the user experience in terms of potential service interruption.

(a) Describe the processes "Break Before Make (BBM) handover" and "Make Before Break (MBB) handover" in the context of cellular networks. Explain the key differences between these two types of handover scenarios and provide examples of when each might occur.**[10 Marks]**

(b) Discuss the roles of base stations and base station controllers in facilitating seamless handover in cellular networks. Highlight the key functions they perform during both intra-BSC and inter-BSC handover scenarios. **[10 Marks]**

Question 5

Antennas are a critical component of wireless networks as they are responsible for transmitting and receiving radio signals between devices. They play a crucial role in determining the range and coverage area of a wireless network.

a) With the help of a diagram, explain how antennas operate by generating and receiving waves. **[10 Marks]**

b) With the help of an example, explain how the size of an antenna is related to the frequency. **[10 Marks]**

(Total: 60 Marks)

---------- **END OF EXAMINATION**----------